

5 (a) (i)

$$\begin{aligned}
 F_E &= \frac{Qq}{4\pi\epsilon_0 r^2} \\
 &= \frac{(3.2 \times 10^{-9})(1.6 \times 10^{-9})}{4\pi\epsilon_0 (0.050)^2} \\
 &= 1.84152 \times 10^{-5} \\
 &= 1.84 \times 10^{-5} \text{ N}
 \end{aligned}$$

(ii)

$$\begin{aligned}
 T \cos 10^\circ &= W \\
 T &= \frac{1.5 \times 10^{-4}}{\cos 10^\circ} = 1.523 \times 10^{-4} \\
 F &= F_E + T \sin 10^\circ \\
 &= 4.48642 \times 10^{-5} \\
 &= 4.49 \times 10^{-5} \text{ N}
 \end{aligned}$$

(iii) The charges in the two spheres will be repelled/redistributed to the two far sides of the sphere.
 The distance between the two charges will larger than 0.050 m and the electric force will be smaller. F will be smaller since the horizontal component of tension remains the same.

(b) The gain in kinetic energy is equal to loss in gravitational potential energy and loss in electric potential energy.

6 (a) As the rod falls, the area enclosed by the rod and the frame decreases and the